

Industrial Design Brief

Battery-Operated Mosquito Repellent Vaporizer · India Market

PROJECT CONTEXT

We are developing a cordless, battery-operated mosquito repellent vaporizer for the Indian market. There is currently no direct competitor in this format in India - plug-in devices (Good Knight, All Out) dominate, but require a wall socket. This product is a portable alternative, inspired by Thermacell (US), built for Indian households, travellers, and outdoor B2B settings like rooftop restaurants and hotels.

PRODUCT FORMAT

Liquid vaporizer using a wick + PTC ceramic heater mechanism. Must be compatible with existing Good Knight / All Out standard liquid refill bottles - this is non-negotiable. No proprietary refill at launch.

POWER - TWO SKUS

V1 - Premium

Rechargeable Li-ion battery with USB-C charging port.

V2 - Value

AA battery operated. Wider distribution, lower cost.

WHAT THE DEVICE MUST HAVE

- PTC ceramic heater - temperature-regulated, safe to touch externally
- Metal wick contact to interface with liquid refill bottle
- ON/OFF button - single control, no additional inputs
- LED indicator - power on/charging status
- Child-safe refill compartment - secured, not easily opened
- Stable flat base - must not tip over on a surface
- Ventilation slots on enclosure for vapour dispersion
- USB-C port (V1) - accessible, not awkwardly placed

DESIGN DIRECTION

Form & Feel

- Compact, upright form - similar footprint to a small desk speaker or candle
- Feels premium in hand - not cheap, not clinical
- Can sit on a dining table, bedside table, or camping surface without looking out of place
- Inspired reference: Thermacell (US) - study the form, improve for Indian context

Material & Finish

- Enclosure: Injection moulded ABS or PP
- CMF to be proposed by agency - we are open to direction
- Avoid glossy surfaces - matte or soft-touch preferred
- Must feel safe for households with children - no sharp edges, no exposed metal

KEY CONSTRAINTS

Manufacturing

- DFM is critical - design must be producible in India or China at scale
- Enclosure target: ~2 steel moulds (body + base)
- Electronics will be designed by a separate PCB engineer - you must share enclosure dimensions with them early

Simplicity

- No display. No app. No smart features.
- One button. One LED. That's it.
- Complexity kills margins and creates compliance headaches - keep it simple

WHAT WE NEED FROM YOU

- Concept sketches / renders (2–3 directions)
- 3D CAD model - SolidWorks or STEP files
- 2D engineering drawings for manufacturing
- CMF specification (colour, material, finish)
- DFM sign-off before tooling
- Coordination with electronics engineer on enclosure dimensions

Note: BIS certification and compliance are being handled separately. The electronics/PCB engineer is a separate engagement. We will coordinate the two teams - please flag any dimension or fit dependencies early.